



Mentoring Fundamental Deep Learning

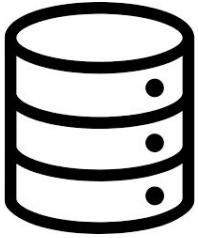
By Johanes Andre R



Rekomendasi Timeline By Dicoding

<https://docs.google.com/spreadsheets/d/1n1RhVwkMApzBbcW4MflNmqdvwy46mYeYRh-fLUB7xKc/edit?usp=sharing>

Machine Learning Step



Data Loading
-Dataset



Data Analytics

- Exploratory Data Analysis - Deskripsi Variabel
- Exploratory Data Analysis - Menangani Missing Value dan Outliers
- Analysis

OVERVIEW

1. DEEPLARNING -> CNN, RNN
2. Neural Network -> Preprocessing, Sekuensial, Plot Loss
3. NLP -> Preprocessing, Ekstraksi Fitur, Classification,
4. **PROYEK PERTAMA**
5. Time Series -> Preprocessing, Matrik,
6. Img Classification ->
7. Recommendation System
8. Conv Model ML
9. Gen AI
10. **Final Submission**

NLP Projects Step:

1. Tokenizer -> Proses mengonversi kata-kata ke dalam bilangan numerik
2. Embedding -> k memahami makna di setiap kata dan mengelompokkan kata-kata dengan makna yang mirip agar berdekatan
3. LSTM -> model dapat memahami makna sebuah kalimat berdasarkan urutan kata
4. One Hot Encoding -> merepresentasikan variabel kategori sebagai nilai numerik
5. Label Classification

Tips Dataset: From Kaggle

 Search



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Your username

johanesandre (not editable)

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5001361

API

Using Kaggle's beta API, you can interact with Competitions and Datasets to download data, make submissions, and more via the command line. [Read the docs](#)



Ensure kaggle.json is in the location ~/.kaggle/kaggle.json to use the API.

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Create New Token

Expire Token

Tips Dataset: From Kaggle

```
17  
18 #ambil API KAGGLE  
19 os.environ['KAGGLE_USERNAME'] = "johanesandre"  
20 os.environ['KAGGLE_KEY'] = "3ed3aa00dd55efd9d337bed2f366d83d"
```

Download Dataset dari Kaggle



```
1 #Download Dataset Kaggle  
2 !kaggle datasets download -d uciml/red-wine-quality-cortez-et-al-2009
```

```
Downloading red-wine-quality-cortez-et-al-2009.zip to /content  
0% 0.00/25.6k [00:00<?, ?B/s]  
100% 25.6k/25.6k [00:00<00:00, 44.0MB/s]
```

<https://www.kaggle.com/datasets/jonathanbesomi/superheroes-nlp-dataset>

Tips Dataset: From Google Drive

Mount Google drive to colab

```
[ ] 1 from google.colab import drive  
    2 drive.mount('/content/drive')
```

Permit this notebook to access your Google Drive files?

This notebook is requesting access to your Google Drive files. Granting access to Google Drive will permit code executed in the notebook to modify files in your Google Drive. Make sure to review notebook code prior to allowing this access.

No thanks

[Connect to Google Drive](#)

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Google Drive for desktop meminta akses ke Akun Google Anda

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- ☒ Melihat foto, video, dan album dalam Google Foto Anda. [Pelajari lebih lanjut](#)
- ☒ Mengambil konfigurasi dan eksperimentasi

Tips Dataset: From Google Drive



```
1 import zipfile
2 with zipfile.ZipFile('/content/drive/My Drive/FM/face-mask-detector.zip', 'r') as zip_ref:
3     zip_ref.extractall('/content/drive/My Drive/')
4
```

Tokenizer



```
1 #Lakukan Tokenizer
2 from tensorflow.keras.preprocessing.text import Tokenizer
3 from tensorflow.keras.preprocessing.sequence import pad_sequences
4
5 tokenizer = Tokenizer(num_words=5000, oov_token='x')
6 tokenizer.fit_on_texts(sinopsis_latih)
7 tokenizer.fit_on_texts(sinopsis_test)
8
9 sekuens_latih = tokenizer.texts_to_sequences(sinopsis_latih)
10 sekuens_test = tokenizer.texts_to_sequences(sinopsis_test)
11
12 padded_latih = pad_sequences(sekuens_latih)
13 padded_test = pad_sequences(sekuens_test)
```



```
1. model = tf.keras.models.Sequential([
2.     tf.keras.layers.Embedding(jumlah_kata, dimensi_embedding, panjang_input),
3.     tf.keras.layers.Flatten(),
4.     tf.keras.layers.Dense(24, activation='relu'),
5.     tf.keras.layers.Dense(1, activation='sigmoid')
6. ])
```

[tf.layers.Dense](#)

-> jumlah perseptron yang dimiliki oleh layer

-> Grid Search :

Refrensi

[Keras.losses model](#)

[Keras.optimizer](#)

CONTOH PROJECT

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